



Pepsi's AR advertisement turns a bus stop wall in London into a fake window
(Credit: www.theverge.com)

images, making users' interaction more interesting and enjoyable. The next development was merging computer graphics with the physical environment and making them interactive in response to users' voice, gestures and touch.

Currently, AR is being used to project graphics onto sports pitches to identify markings and distance, and promote products and brands. It is increasingly becoming an indispensable market tool for business houses to effectively engage shoppers with their product and brand story. **The AR industry is a steadily growing market and slated to touch almost US\$ 120 billion by 2020 with one billion people estimated to be using it.**

AR is an experience that involves integration of digital information with information from the user's real-world environment on real-time basis. This experience is usually displayed on a screen or monitor, which could be as small as a tablet or mobile, or as large as an advertising billboard. AR uses visuals, sound and even haptics (touch/movement) to create an immersive experience. Simply put, AR is an embellished or altered form of reality where content lays over the user's real-world views, enhancing information about the world around.

While shopping, extra information about products can be obtained by just scanning the product tag with a smartphone. With AR, it becomes possible

to try out clothes and accessories in the comfort of your own home, instead of physically going to the marketplace.

What is VR

The concept of VR dates back to the 1930s, when Stanley G. Weinbaum, a science fiction writer, in his book *Pygmalion's Spectacles* referred to a game in which players wear goggles to watch a holographic recording of virtual stories, and experience touch and smell.

The real development started in 1968, when Ivan Sutherland, an electrical engineer at Harvard University, created a head-mounted display (HMD) system for military training applications, nicknamed The Sword of Damocles. The massive headset, being very heavy, was anchored to the ceiling and the user was strapped into the system, making the whole experience quite uncomfortable.

In 1990s, VR headsets were used in some arcade games for simulation. The first home system, Virtual Boy from Nintendo, appeared in 1995. It had stereo sound, LCD screens and head tracking. It did not become a big hit as using the console was uncomfortable and the software was not easily available.

A revolution in VR technology took place in 2014 when Facebook acquired Oculus VR system. Oculus introduced modern easy-to-use plug-and-play compatible headsets with a

wide range of supported applications. Modern VR devices are available with 360° cameras and cheaper headsets, and improvements are continuously taking place to improve 3D graphics and make the devices wireless and faster. VR training tools are being widely used in military for combat situations, flight exercises and the like.

VR is an artificial, computer-generated simulation or recreation of a real-life environment or situation. It makes users experience simulated reality first-hand using vision and hearing. It transforms users into a 360-degree environment where they can move around at their own pace to explore whatever is of interest.

A headset is necessary to experience VR. Mostly, VR is used in gaming, entertainment, simulators, education and marketing industries, and is an excellent tool for practising medicine, surgery, flying, driving and so on. The user wears an HMD, makes simple head movements and looks around to interact with the virtual world.

Difference between AR and VR

Both VR and AR present users an enhanced or enriched experience. **VR is a distinct concept from AR, as there is no element of real world involved, whereas AR is the blending of computer-generated elements with the real world.** AR adds virtual components such as digital images and graphics to perceive a live view with the help of a camera. It may or may not interact with the elements in the live view.

When using VR, what one sees and experiences is different from what is actually around the user, since it is entirely simulated. AR is less intrusive, costs less and is easier to apply to everyday life, versus VR.

Use of AR and VR as a marketing tool

Many brands have either adopted or are thinking of adopting creative ways to promote their products.

Digital technology for advertising conventionally uses online video, pop-ups, pop-unders, displays, banners, etc. Several users found these annoying and intrusive. As a result, blockers started appearing, and this reduced the effectiveness of the advertise-